Two literal projective-plane constructions for $\mathrm{PSL}_3(4)$

A bounded recovery of constructions 5.1 and 5.2 in Gottschalk–Leemans

The reference edition is the complete 22-page Halle preprint [1], not an uncollated later journal edition. We recover the two incidence constructions on its physical pages 14–16, Lemmas 4.1–4.3 and Table 1. The finite certificate also treats the specified truncations below. This is an existence and incidence result, not an exhaustion of higher-rank geometries. Automorphism and correlation groups, conjugacy-fusion counts, doubling and the other constructions are not asserted here.

The actual finite model. Put $F = \mathbb{F}_2[w]/(w^2 + w + 1)$ and label its elements $0, 1, 2 = w, 3 = w+1$. List the 21 projective points by lexicographically sorting the nonzero column vectors whose first nonzero coordinate is 1. Lines are the kernels of the 21 normalized nonzero linear forms. A hyperoval is a six-subset with no three collinear points. Let $G = \mathrm{PSL}_3(4)$ in its point action.

We use the two actual degree-21 generators retained in the independently reviewed rank-two component. No equality of independently ordered labels is presumed. A five-point orbit of its order-960 line stabilizer gives a line; the orbit of that set gives all 21 lines. The certificate verifies the coordinate-to-retained label map

$$(6, 2, 10, 14, 18, 1, 7, 8, 9, 3, 11, 16, 21, 4, 19, 12, 17, 5, 15, 20, 13).$$

It checks every line under this bijection. In these coordinates the actual two retained generators are represented, respectively, by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & w & 0 \\ 0 & 0 & w+1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}. \tag{1}$$

Both determinants are 1; all 21 images are checked, not merely a projective frame. Closure of the two literal permutations has 20,160 elements. This is also $|\mathrm{SL}_3(4)|/3 = 4^3(4^3 - 1)(4^2 - 1)/3$, so the realization is the full named projective special linear group. The retained named-model identification and finite-field conventions are explicit inputs; the literal matrix and permutation checks additionally verify their use here.

Testing all $\binom{21}{6} = 54,264$ subsets gives 168 hyperovals, in three G-orbits of 56. Fix distinct orbits $\mathcal{H}_3, \mathcal{H}_4$. The five types are lines, points, $\mathcal{H}_3$, $\mathcal{H}_4$, and unordered point pairs. The type sizes are $(21, 21, 56, 56, 210)$. For $h \in \mathcal{H}_3$, $k \in \mathcal{H}_4$ and Q a point pair, define the ten relations by

	p	h	k	Q
ℓ	$p \notin \ell$	$\ell \cap h = \varnothing$	$\ell \cap k \neq \varnothing$	$\ell \cap Q = \varnothing$
p		$p \in h$	$p \in k$	$p \in Q$
h			$\lvert h \cap k \rvert = 3$	$Q \subset h$
k				$Q \subset k$.

(2)

This is $\Gamma_{5.1}$. For $\Gamma_{5.2}$ replace only $\ell \cap k \neq \varnothing$ by $\ell \cap k = \varnothing$. These are the literal rules in the source; in particular line–point incidence is nonincidence in the underlying projective plane.

Theorem 1. *Each of these two incidence systems is a firm rank-five geometry with 20,160 chambers. The actual G acts regularly on chambers and transitively on flags of every type. Each geometry is residually connected, residually weakly primitive, and satisfies* IP_2. *Its rank-one residue actions are doubly transitive. For $\Gamma_{5.1}$ all twenty one-type extension numbers in source Table 1 are those displayed in Table 1. The choice of the ordered pair of distinct hyperoval orbits does not change either geometry up to a type-preserving isomorphism normalizing the specified G-action.*

fixed types	added type	number	fixed types	added type	number
123	4	5	123	5	5
124	3	3	124	5	3
125	3	2	125	4	2
134	2	3	134	5	3
135	2	2	135	4	2
145	2	2	145	3	2
234	1	3	234	5	2
235	1	6	235	4	4
245	1	6	245	3	4
345	1	3	345	2	2

Table 1: Exact source Table 1 extension numbers for $\Gamma_{5.1}$.

Exact finite verification and its completeness. The accompanying program uses only integer masks, permutations and finite field arithmetic after exporting the fixed input generators and line orbit. For a prescribed ordered type set I, it recursively chooses an object of each type, intersecting the remaining allowable sets with its adjacency row. Induction on $|I|$ proves that the returned tuples are precisely all flags of those types, without repetition. This procedure is applied to every subset of the five types, including the empty subset.

Let $\mathcal{C}$ be the resulting complete set of full flags and choose $C \in \mathcal{C}$. Applying every one of the 20,160 literal elements of G to C gives 20,160 distinct tuples, exactly $\mathcal{C}$. All generator actions preserve every object set and all ten incidence relations. Thus this is a genuine regular chamber action; its kernel is trivial, since an element in the kernel fixes C. Independently, for every type subset I the complete flag list is compared with the projection of $\mathcal{C}$ onto I. Equality proves that every partial flag extends to a chamber; chamber transitivity alone is not substituted for this check. It then also proves flag transitivity.

For each $F \subseteq C$, the residue objects of type i are computed as the intersection of the adjacency rows of the objects of F. Its action uses precisely those literal elements of G fixing every object of F. No permutation-group quotient or replacement action is used. The incidence graph of each residue of rank at least two is connected by direct traversal. Each panel has at least two extensions, proving firmness under the standard panel convention; a chamber itself is not required to have two extensions. Because every flag is conjugate to the corresponding subflag of C, these are checks of all residues, including those based at flags not chosen in the calculation.

Lemma 1. *Let K act transitively on a finite set X, with $x \in X$. For one representative y of each K_x-orbit on $X \setminus \{x\}$, form the undirected graph with edges $\{x^g, y^g\}$ for all $g \in K$. The action is primitive if and only if all these graphs are connected.*

Proof. Connected components of any such graph are a K-invariant equivalence relation, so a disconnected graph contradicts primitivity. Conversely, if a nontrivial block contains x and $y \neq x$, all edges in that pair orbit lie within blocks; its graph cannot be connected. Choosing another point in the same K_x-orbit gives the same graph. □

The certificate applies this lemma to the actual actions of each G_F on each remaining type. Every residue has at least one primitive type action, precisely the weak primitivity required by RWPRI. Double transitivity is recorded separately by the orbits of an actual point stabilizer; it is never a filter. For a rank-two residue, the program checks that the graph is complete bipartite or that each pair on one side has at most one common neighbour. The latter is symmetric (a pair of common neighbours would give a quadrangle) and proves the partial-linear condition IP_2. These checks use every pair of local objects.

The finite output retains all 32 flag counts, all actual stabilizer orders, all 31 positive-rank residue records, every local primitivity and double-transitivity flag, and every rank-two test for

each model. In both models the panel orders in type order are $(2,1,1,1,1)$; equivalently the five panel sizes are $(3,2,2,2,2)$. Their chamber stabilizers are trivial. Table 1 is computed directly at the corresponding base triple flags; the already verified flag action transports it to every triple flag.

The six choices of hyperoval orbits. The coordinate maps $\mathrm{diag}(w,1,1)$ and $(x,y,z) \mapsto (x^2,y^2,z^2)$ preserve the plane and normalize the actual group: conjugates of both retained generators are checked in the complete literal G. Their permutations of the three hyperoval orbits are a 3-cycle and a transposition. They therefore act transitively on the six ordered pairs of distinct orbits. For each of these six pairs, for both models, and for each of the two maps, the certificate stores the resulting bijections on all five object types and checks every incidence and nonincidence. These 24 complete maps pay the choice-independence assertion. They are not a computation of the full automorphism group.

The specified truncations. The literal restrictions in Table 2 have the indicated source profiles. Here the type numbers always refer to (2), B is the chamber stabilizer in the specified G, and panel orders are listed in increasing retained-type order. For each row the program again enumerates every partial flag and chamber, proves flag transitivity, and verifies every base residue, firmness, RWPRI and IP_2. It checks that projection of all rank-five chambers is exactly the truncated chamber set.

source label	parent	retained types	panel orders	$\lvert B\rvert$
4.1	5.1, 5.2	2345	$(1,3,3,1)$	3
4.10	5.1	1345	$(2,1,1,2)$	2
3.1	5.1, 5.2	235	$(1,3,4)$	12
3.6	5.1	345	$(3,3,2)$	6
2.2	5.1, 5.2	35	$(3,14)$	24

Table 2: Eight checked literal truncation routes.

The object maps are the identity on each retained type; incidence is literally restricted, rather than inferred from the numerical profile. The two 4.1 restrictions and the two 3.1 restrictions are therefore identical incidence models: they omit the only relation changed between 5.1 and 5.2. In particular the former also restricts to the latter and to 3.6. The 4.10 restriction likewise restricts to 3.6. These maps establish the advertised construction routes in this fixed edition. They do not identify an arbitrary geometry from its diagram alone. No further independent realization from the cited earlier papers is being asserted isomorphic without a supplied map.

The rank-two 2.2 restriction has local degrees $(15,4)$ with parabolic orders $(360,96)$ and intersection order 24. Its actual local double-transitivity flags are $(\mathrm{false},\mathrm{true})$, consistent with the corrected accepted rank-two component: $14 \nmid 24$. Thus the local-2T distinction is preserved through the higher-rank construction.

Scope and attribution. The constructions and their intended geometric properties are due to Gottschalk and Leemans and to the predecessors explicitly cited in their source. The new material here is a transparent finite certificate of those specified constructions, together with the exact action and source joins. The accepted rank-two model is acquired unchanged. Complete higher-rank classification, other incidence constructions, doubling, full group names and kernels beyond the orders stated above, full automorphism and correlation groups, and fusion multiplicities remain separate obligations. No novelty, whole-source, question, Lean or measurement credit is claimed.

References

[1] H. Gottschalk and D. Leemans, *Geometries for the group* PSL(3, 4), Halle Reports on Algebra and Geometry 99–30, fixed 22-page preprint; physical pp. 8–16 for the source routes used here.

[2] *Primitive residues for* $\mathrm{PSL}_3(4)$, accepted L100 rank-two component, edition 2, with distinct review and root adoption; exact immutable manifests are listed in the certificate dependencies.